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Antifouling Immunomodulatory Copolymer Architectures That Inhibit the Fibrosis of Implants

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Immune reactions to medical implants often lead to encapsulation by fibrotic tissue and impaired device function. This process is thought to initiate by protein adsorption, which enables immune cells to attach and mount an inflammatory response. Previously, several antifibrotic materials have been either designed to reduce protein adsorption or discovered via high-throughput screens (HTS) to favorably regulate inflammation. The present work introduces antifouling immunomodulatory (AIM) copolymer coatings, which combine both strategies to effectively enhance implant protection. AIM copolymers synergistically integrate zwitterionic moieties to resist protein fouling, HTS-derived antifibrotics for immunomodulation, and silane monomers for grafting to diverse substrates including elastomers, ceramics, and metals. Interestingly, simply combining these monomers into conventional random or block copolymer architectures yielded no significant advantage over homopolymers. By contrast, an unusual polymer chain architecture — a zwitterionic block flanked by a mixed zwitterionic immunomodulatory segment — showed superior fibrosis resistance in both peritoneal and subcutaneous sites over one month in immunocompetent mice. This architecture also improved the performance of two different HTS-derived antifibrotic monomers, suggesting that tailoring AIM architectures may broadly complement immunomodulatory chemistries and provide a versatile approach to improving implant longevity.

1. Introduction

The foreign body response (FBR) initiated by the host immune system represents a major hurdle for the long-term function of implants.[1] FBR culminates in a dense collagenous capsule formed around the implant in a process termed fibrosis, hampering the release of therapeutics from implantable drug delivery systems,[2–4] limiting nutrient diffusion into cell encapsulation devices,[5–8] dulling electrical signals from electrode-containing devices,[9,10] and diminishing the sensitivity and accuracy of real-time biosensors.[11,12] Surrounding cells can also initiate a concerted mechanical constriction of the capsular tissue leading to the destruction of devices fabricated using compliant materials such as reconstructive soft tissue implants.[13,14] Moreover, adhesion of fibrous capsules to native tissues and organs can complicate surgical extraction of devices, leading to severe clinical complications.[15–17] The failure and concomitant surgical

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replacement of such devices not only impact patient comfort, quality of care, and financials, but also contribute considerable burden on the healthcare system.

While a complete understanding of fibrosis remains elusive, it is generally thought to begin with non-specific protein adsorption immediately following implantation.[18,19] These proteins facilitate immune cell attachment, which then induce a cascade of pro- and anti-inflammatory cues that attract and mobilize additional cell types such as granulocytes, lymphocytes, and fibroblasts, ultimately leading to excessive collagen deposition.[18,20] Macrophages in particular play a pivotal role in the immune recognition of foreign bodies, orchestration of the immunological cascade, and fusion into foreign body giant cells that line the fibrotic encapsulation and amplify the response.[21,22]

Early efforts to prevent implant fibrosis include administration of immunosuppressive drugs like dexamethasone.[23,24] Systemic immunosuppression, however, leaves patients susceptible to infectious diseases and a range of additional medical complications.[25,26] Localized strategies leveraging controlled release systems lower dose requirements and minimize global immunosuppression, but still face potential challenges regarding exhaustion of the drug depot and incidental systemic immunosuppression.[27–30] Moreover, many of these strategies require dissolved drug to be physically absorbed into the implant or drug particles to be mixed into the material before forming it into a medical device, limiting compatibility.

Monolithic materials with anti-fibrotic properties have emerged as attractive alternatives that forgo immunosuppression. Two conventional design approaches target either upstream non-specific protein adsorption (i.e., fouling), or downstream inflammatory events.[31–33] For example, ultrahydrophilic materials designed to prevent protein fouling strongly interact with water molecules, reducing interactions with ambient proteins.[34–38] Common choices include polyethylene glycol (PEG)[39] or zwitterionic groups such as carboxybetaine,[40] sulfobetaine,[36] and phosphatidylcholine,[6] which form even stronger hydration shells than PEG.[37] In contrast, a distinct class of synthetic antifibrotics has been discovered via high-throughput chemical screens[41–43] (HTS), whose members appear to achieve their function with minimal effect on protein adsorption.[41,44] Despite some studies suggesting that hydrophobic materials may cause greater inflammation,[45] both hydrophobic and hydrophilic lead candidates from these screens have shown promise in reducing fibrosis in rodents and non-human primates when modified on various implant materials.[6,7,41,43,44,46,47] Although their exact mechanisms of action remain unclear, studies indicate that these synthetic antifibrotics elicit pronounced changes in immunological responses relevant to inflammation and tissue remodeling.[6,41,44,48]

Currently, fouling-resistant zwitterions and immunomodulatory HTS-derived chemistries are considered among the most promising chemical modifications for reducing implant fibrosis.[33,49] In contrast to previous efforts to identify promising functional groups, we aimed to synergize current state-of-the-art antifibrotic chemistries by integrating both antifouling and immune-modulation strategies. To achieve this, we developed antifouling immunomodulatory (AIM) copolymers by combining disparate zwitterionic and synthetic immunomodulatory monomers into unique polymer chain architectures. Initially, we synthesized five distinct AIM architectures and evaluated their antifibrotic performance in the peritoneal cavity of immunocompetent mice. Standard random copolymerization and block copolymerization did not yield significant improvements. However, our lead architecture (AIM-5), which integrates a continuous zwitterionic segment with a random copolymer tip, outperformed homopolymers of its constituents. In vitro and in vivo adsorption studies highlight the favorable protein, phospholipid, and cellular binding profiles of AIM-5. RNA sequencing and cytokine analysis of primary macrophages reveal that AIM-5 surfaces elicit a balanced immune response. Finally, we validated the efficacy and versatility of this architecture by synthesizing an AIM-5 analog using an additional bioactive moiety and deploying it in the subcutaneous space as an alternative implantation site. These findings motivate future efforts to develop new copolymer architectures to complement existing efforts to discover antifibrotic small molecule moieties.

2. Results

2.1. Design and Synthesis of Antifouling ImmunoModulatory (AIM) Copolymers

AIM copolymer coatings combine functional monomers derived from two different classes of antifibrotic chemistries, along with a reactive handle for grafting to implant surfaces (**Figure 1A**). The first type of monomer contains a super-hydrophilic zwitterionic side group that confers resistance to biomolecule fouling due to its strong preferential interaction with water molecules.[37] The other monomer originates from libraries of chemical modifications on alginate hydrogels which were screened to identify moieties with antifibrotic properties.[41,43] These relatively hydrophobic modifications have limited antifouling properties but appear to reduce the buildup and activation of cell types associated with the fibrosis of implants.[6,44,48] A reactive alkoxy silane monomer enables AIM polymers to covalently graft onto a diverse range of substrates via silanol condensation.[50] For the antifouling monomer, we utilized methacryloyloxyethyl phosphorylcholine (MPC), which we previously found to be the most effective zwitterionic group for reducing the fibrotic accumulation on the surface of implants.[6] For the immunomodulatory monomer, we

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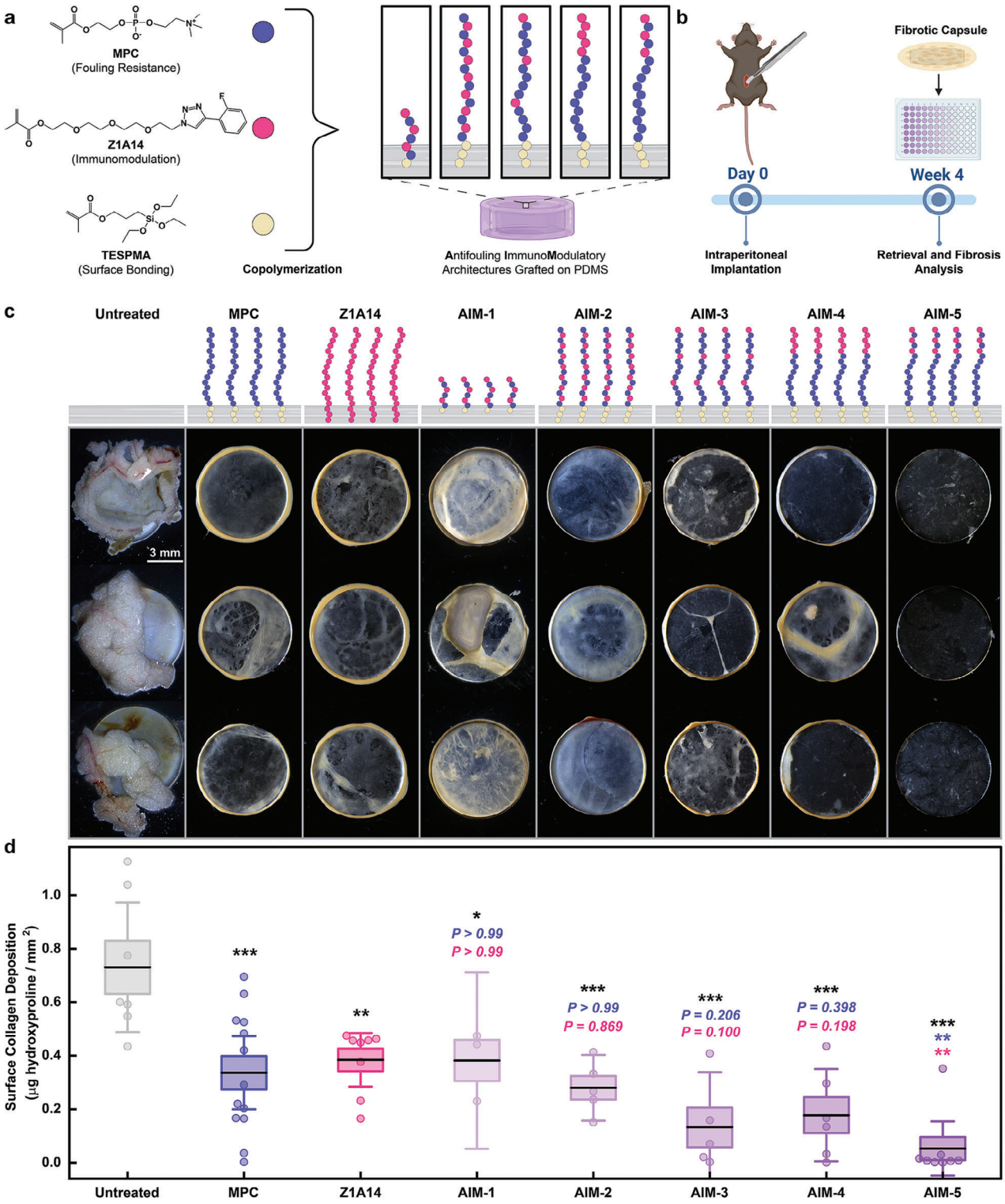


Figure 1. Design and testing of Antifouling ImmunoModulatory (AIM) copolymer architectures. A) Schematic of AIM polymer designs and the chemical structures of constituent monomers. B) Schematic overview of the animal experiment used to characterize antifibrotic properties. C) Ball model illustrations of the different coatings tested (top) and vertically stacked optical micrographs (bottom) of three different explanted samples from each group after 4 weeks of implantation. D) Biochemical quantification of surface collagen density determined by hydroxyproline assay. Black central lines denote the mean value, boxes indicate standard error, and whiskers represent the 95% confidence interval. Statistical significance was determined from Dunnett's multiple comparison tests. The color of the significance markers indicates comparisons made relative to untreated (black), MPC (blue), or Z1A14 (pink). Significance markers follow: $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$.

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synthesized monomers containing either fluorobenzene triazole (Z1A14)[43] or tetrahydropyran triazole (Z2Y12)[41] (Figures S1–S3, Supporting Information), which represent lead candidates from 2 different chemical screens of antifibrotic moieties. Triethoxysilyl propyl methacrylate (TESPMA) acts as a surface-grafting group which we found to offer comparable in vivo performance to other grafting methods, including click chemistry and graft-from polymerization, in a preliminary investigation of the effect of grafting method on the antifibrotic capabilities of MPC homopolymers (Figure S4, Supporting Information).

Due to the disparate physicochemical properties of the functional monomers, we hypothesized that the sequence in which the monomers are copolymerized might affect the surface hydrophilicity and immunomodulatory side group presentation. Specifically, we postulated that maintaining continuous stretches of zwitterionic functional groups will minimize hydrophobic interruptions in the ordered hydration layer, thereby improving repulsion of profibrotic foulants. To synthesize AIM copolymers with variable molecular architecture, we utilized photo-induced atom transfer radical polymerization (photoATRP) which enables the generation of copolymers with definable block lengths (Figure S5, Supporting Information).[51] Initially, we polymerized MPC homopolymers. Time-series nuclear magnetic resonance spectroscopy (NMR) measurements taken throughout the polymerization reveal first-order reaction kinetics with respect to the monomer, indicating a constant concentration of active radical centers (Figure S6, Supporting Information). This ensures that polymer chains may be reinitiated for chain extension by other monomers. Importantly, linear kinetics are maintained by both MPC and Z1A14 when polymerized together to form AIM copolymers (Figure S7, Supporting Information). The incorporation of the two monomers in copolymer products was apparent by both proton and fluorine NMR (Figure S8, Supporting Information). Extension with TESPMA provided surface grafting capabilities, with MPC and AIM polysilanes remaining stable in ethanolic solutions for several months (Figure S9, Supporting Information). Using this approach we successfully synthesized 5 different AIM copolymer architectures with varying molecular weight, molar ratio of MPC to Z1A14, and monomer arrangement. The first three AIM architectures are random copolymers of the zwitterionic and immunomodulatory monomers. AIM-1 is a short (≈12 kDa) copolymer with a 1:1 ratio of MPC to Z1A14. AIM-2 is a long (≈41 kDa) copolymer also with a 1:1 monomer ratio. AIM-3 is a long (≈44 kDa) copolymer with a greater MPC to Z1A14 ratio of 4:1. The other two architectures feature a continuous block of zwitterionic MPC monomer flanked by an immunomodulatory tip segment. AIM-4 has a pure tip of Z1A14 while AIM-5 has a 1:1 random copolymer tip of Z1A14 and MPC. All AIM polymer architectures feature a TESPMA block for grafting onto implant surfaces. Ball model representations of these architectures are schematically illustrated in Figure 1A, and a summary of the reaction ratios for each polymerization step can be found in Table S1 (Supporting Information).

To assess the effect of AIM polymers we first applied the polymers as coatings on polydimethylsiloxane (PDMS), a common elastomer used in a wide range of medical implants including catheters, prostheses, drug delivery systems, and bioelectronics.[52] The coating procedure involved activating the PDMS surface via air plasma treatment, followed by immersion in a 0.5 wt.% solution of either MPC or AIM polymer in ethanol with 5 vol% water added to hydrolyze the alkoxy silane block. Due to the hydrophobicity of Z1A14, polymeric Z1A14 had poor solubility in alcohols, even at low degrees of polymerization. Consequently, as a control, we treated plasma-activated PDMS with Z1A14 following previous reports[43,53] to induce surface-initiated free radical polymerization.[54–56] After thorough washing and annealing, X-ray photoelectron spectroscopy confirmed successful coating on PDMS surfaces by showing the presence of nitrogen, phosphorus, and fluorine (Table S2, Supporting Information). As a proof-of-concept, we also validated that the versatile polysilane grafting chemistry is generalizable to alternative substrates including glass and Ti-6Al-4 V, a titanium alloy commonly used for medical implants[57] (Figure S10 and Table S3, Supporting Information). Water contact angle measurements showed that all AIM modifications significantly increase the hydrophilicity of PDMS, with AIM-5 achieving a contact angle comparable to that of MPC homopolymer (Figure S11, Supporting Information). Although hydrophobic recovery is characteristic of PDMS,[58] measurements taken after 8 months of storage in sterile water at 4 °C indicate only minimal changes in hydrophilicity (Figure S12A, Supporting Information). Furthermore, polysilane coatings feature even greater stability on non-elastomeric substrates, as evidenced by measurements on Ti-6Al-4 V after 5 months of storage under ambient conditions, which show no appreciable degradation in surface properties (Figure S12B, Supporting Information). All coatings on PDMS appeared optically transparent; however, scanning electron microscopy of coated surfaces revealed nanoscale precipitates in Z1A14-coated samples (Figure S13, Supporting Information). The hydrophilic MPC and AIM polymers did not have any observable precipitates but exhibited minor rippling on the surface, likely due to differential swelling during drying.

2.2. AIM Polymer Architecture Modulates Fibrosis of Implant

Coated PDMS disks (diameter = 8 mm, thickness = 1.5 mm) were implanted into the peritoneal cavity of immunocompetent C57BL/6 mice for 4 weeks before retrieval and analysis of fibrotic tissue (Figure 1B). Previous investigations suggest that fibrosis peaks in the initial weeks post-implantation before receding and plateauing in the following months,[43,59] making 4 weeks a suitable period to assess changes in fibrosis at its maximum. Figure 1C presents optical micrographs of explanted samples from each group, and Figure 1D displays biochemical quantification of the collagen content on each implant. MPC and Z1A14 samples show similar efficacy at reducing the fibrotic build-up on implants relative to control, which aligns with prior observations comparing lead candidates from combinatorial chemistry screens against pure zwitterionic coatings.[6,43] Interestingly, while all AIM coatings also conferred a significant reduction in fibrosis relative to control, only AIM-5 significantly outperformed both MPC and Z1A14 (Figure 1D). Connective tissue staining on histological sections mirror these findings, highlighting a drastic decrease in both attached cells and collagenous tissue on the surface of AIM-5 samples (Figure S14, Supporting Information).

A comparative analysis of the performance of different AIM architectures offers insights into the effects of molecular weight,

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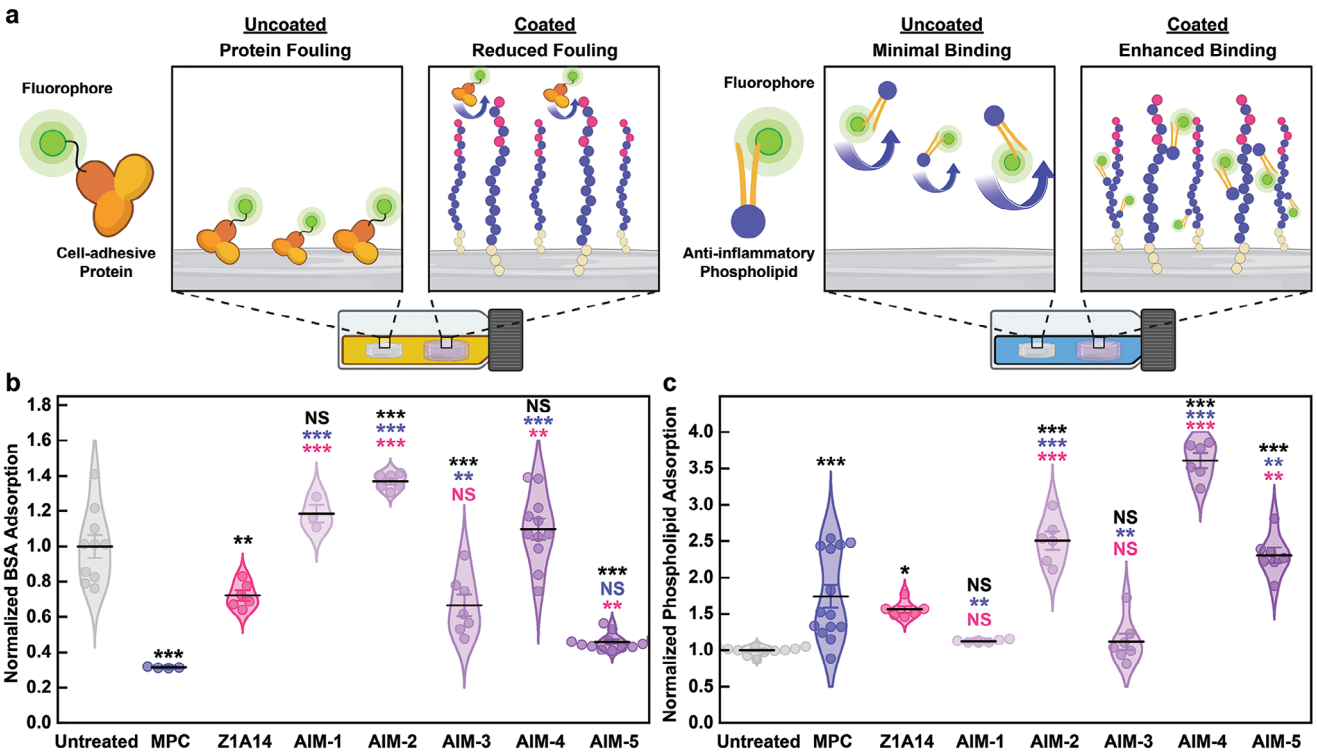


Figure 2. Evaluation of biomolecular adsorption on AIM polymer surfaces. A) Schematic of the biochemical binding assays using fluorescently tagged albumin and phosphatidylcholine as models for profibrotic proteins and anti-inflammatory phospholipids respectively. Super hydrophilic phosphocholine zwitterions act to repel protein foulants, while both phosphocholine and immunomodulatory moieties improve affinity to phospholipids. B) Violin plots quantifying the adsorbed albumin protein or C) phosphatidylcholine based on fluorescence intensity relative to untreated surfaces. Violin contours result from kernel density estimations using Scott's method for bandwidth selection, error bars indicate standard error, and the black central lines denote the mean. Statistical significance was determined from Dunnett's multiple comparison tests. The color of the significance markers indicates comparisons made relative to untreated (black), MPC (blue), or Z1A14 (pink). Significance markers follow: $*P < 0.05$, $**P < 0.01$, $***P < 0.001$.

monomer ratios, and monomer organization. AIM-1 (12.3 kDa) and AIM-2 (41.3 kDa), random copolymers synthesized with equimolar ratio of MPC and Z1A14 but molecular weights that differ by several fold, performed close to MPC and Z1A14 alone. AIM-3, with similar molecular weight to AIM-2 but a higher MPC to Z1A14 ratio (4:1) showed reduced collagen deposition. AIM-4, featuring segregated blocks of MPC and Z1A14, had comparable performance to AIM-3. AIM-5, comprising a continuous block of MPC at its base and a cap formed by an equimolar random copolymer of MPC and Z1A14, exhibited near complete inhibition of implant fibrosis. Taken together, these results indicate that the architecture of AIM polymers has a substantial effect on the fibrotic response.

2.3. AIM Polymers Reduce Protein Adsorption and Enhance Anti-Inflammatory Phospholipid Binding In Vitro

To probe possible mechanisms by which AIM polymers derive their antifibrotic properties, we conducted a series of biomolecular adsorption studies. Protein adsorption initiates fibrosis by enabling host cell attachment and triggering inflammation.[18,20,60] Conversely, certain classes of phospholipids such as phosphatidylcholine, phosphatidylethanolamine, and sphingomyelin exhibit anti-inflammatory effects and protect against fibrosis in multiple organs.[61–64] Additionally, it was shown that multiple top performing antifibrotic chemistries from high-throughput screens preferentially bind such anti-inflammatory phospholipids in vivo.[53] Zwitterionic materials like MPC are well known to reduce protein adsorption,[37,65] but their effects on phospholipid interactions are less clear, with different studies reporting either improved binding[66–68] or repulsion.[69] Here, we studied how the molecular sequence of Z1A14 and MPC moieties impact protein and anti-inflammatory phospholipid interactions.

Adsorption studies involved incubating PDMS samples with fluorescently tagged versions of model cell-adhesive proteins and anti-inflammatory phospholipids. Specifically, we used bovine serum albumin (BSA), the most abundant protein in plasma,[70] and phosphatidylcholine, the most abundant phospholipid in cellular membranes[62] and pulmonary surfactant.[71] After incubation and thorough rinsing, we imaged samples to evaluate their relative fluorescence intensity as a measure of adsorbed analyte (**Figure 2A**).

Figure 2B reveals that although all coatings increased the hydrophilicity of PDMS surfaces (Figure S11, Supporting Information), AIM-1 and AIM-4 did not significantly reduce protein fouling, and AIM-2 even increased protein adsorption by nearly 40% relative to untreated samples. On the other, MPC and AIM-5 both reduced fluorescent BSA signal by over 2-fold, while Z1A14 and AIM-3 provided a more modest reduction of ≈30%. The robust

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antifouling behavior of AIM-5 and MPC aligns with their extremely hydrophilic properties, characterized by contact angles well below the so-called Berg limit (≈50–60°),[72] below which surfaces begin to repel proteins. By comparison, the other antifibrotic coatings fall above this threshold and exhibit variable fouling behavior, consistent with other studies on moderately hydrophilic surfaces[45]

Phospholipid binding data generally shows an opposite trend across groups (Figure 2C). Most coatings increased phospholipid attachment, with MPC and Z1A14 coatings enhancing it by ≈50%. AIM-2, AIM-4, and AIM-5 architectures displayed the highest affinities (>200% relative to control) and significantly outperformed both MPC and Z1A14 alone.

The variable properties of these copolymer architectures may partially be explained by conformational differences driven by their sequence of disparate monomers, similar to how protein folding and concomitant function depend on the specific sequence of amino acids with variable hydrophilicity. Although all five AIM copolymers significantly protected against implant fibrosis in our mice studies (Figure 1C,D), their binding patterns varied considerably. All AIM architectures, except AIM-1, exhibited at least one favorable property: low BSA binding or high phospholipid affinity. This suggests varying mechanisms for their antifibrotic effects. For example, AIM-2 and AIM-4, with poor resistance to cell adhesive BSA but with high phospholipid affinity, likely rely more on anti-inflammatory pathways. AIM-3, with a greater MPC content, had the opposite interaction profile. AIM-5, which showed the greatest in vivo fibrosis resistance, had a uniquely favorable adsorption profile, matching MPC's efficacy in protein reduction and outperforming both MPC and Z1A14 in anti-inflammatory phospholipid binding.

2.4. Lead AIM Architecture Reduces Cellular Adhesion and Protein Fouling In Vivo

Building on our in vitro adsorption studies, we aimed to evaluate the antifouling properties of our lead AIM architecture in vivo. Here, we investigated the immediate response to PDMS implants placed subcutaneously in C57BL/6 mice for 30 min. After retrieval, we stained samples to analyze the number of attached cells and the presence of IgG and fibronectin, proteins classically implicated in cellular adhesion.[60,73,74]

Figure 3A presents confocal microscope images of samples with different surface treatments and stains. Nuclear staining reveals that Z1A14 reduced the density of surface-attached cells by ≈50%, while MPC and AIM-5 treatments led to a near-complete reduction in attached cells (Figure 3B). Z1A14 samples experienced similar levels of IgG and fibronectin deposition to that of untreated samples, with dense, fibrous patches of proteins covering over 75% of the implant surface area (Figure 3C). In contrast, MPC and AIM-5 samples showed overall lower levels of IgG, with more uniform protein density in the regions where observed. Fibronectin, however, was even less detectable on the surfaces of MPC and AIM-5 samples. These observations may be attributed to the greater abundance and lower molecular weight of IgG (≈150 kDa) compared to fibronectin (≈250 kDa (monomeric) – 1000 kDa (tetrameric)), making it more readily adsorbed onto implants as dictated by the Vroman effect.[75] Combined with results from our in vitro adsorption studies, these data indicate that the AIM-5 architecture effectively integrates hydrophobic immunomodulatory moieties with zwitterionic moieties in manner that retains potent antifouling properties.

2.5. AIM Polymer Surfaces Modulate the Immunological Response of Primary Macrophages

After examining the antifouling properties of our lead AIM copolymer, we turned our attention toward its immunomodulatory function. We cultured primary bone marrow-derived macrophages from C57BL/6 mice on PDMS substrates modified with Z1A14, MPC, or AIM-5. Following two days of incubation, we performed bulk RNA sequencing and quantified a panel of secreted cytokines in the supernatant culture media using a multiplex bead-based immunoassay.

Volcano plots highlighting differentially expressed genes relative to macrophages cultured on pristine PDMS indicate that both Z1A14 and AIM-5 generate more pronounced transcriptomic modulation than MPC samples (**Figure 4A**). To better understand the relationships between the different groups, we calculated pair-wise Euclidean distances between each sample, treating each gene as a unique dimension, and determined the k-nearest neighbors for each data point. Additionally, we calculated the average inverse inter-group distance as a metric of group similarity. Examining both the UMAP and similarity matrix, we find the most distinct clustering within untreated and Z1A14 samples, with the greatest inter-group similarity between Z1A14 and AIM-5 (Figure 4B and C). Gene set enrichment analysis (GSEA) network plots further corroborate the similarity between Z1A14 and AIM-5 by displaying connections between significantly enriched gene sets, related by the number of overlapping genes across each set (Figures S15 and S16, Supporting Information).

In a more targeted analysis of the biomaterial response, we filtered differentially expressed genes for relevance to antigen presentation, cellular recruitment, tissue remodeling, and immune cell activation. The resulting heatmap in Figure 4D reveals that Z1A14 and AIM-5 downregulate MHC Class II (H2Aa and H2Ab1), which are upregulated in classically activated macrophages.[76] Interestingly, AIM-5 uniquely upregulated MHC Class I expression (H2-K2). All treatment groups increased the expression of chemotactic cytokines. Z1A14 and AIM-5 increased the expression of at least one matrix metalloprotease (MMP9 and MMP12, respectively), which control fibrosis by regulating extracellular matrix (ECM) composition and macrophage phenotype.[77–80] Likewise both immunomodulatory surfaces, but not MPC, prompted significant changes in various extracellular matrix components including collagens and laminin. Z1A14 and AIM-5 also uniquely downregulated several clusters of differentiation. Downregulation of CD74 is particularly noteworthy as previous studies on another lead antifibrotic moiety, Z2Y12, established that blocking CD74 receptors on macrophages reduces the immunomodulatory action of the biomaterial surface.[48] Furthermore, AIM-5 and Z1A14 significantly perturbed several additional markers for inflammation, including leukocyte immunoglobulin-like receptors (Lilra6, Lilrb4a, Lilrb4b), arginase (Arg1), tumor necrosis factor, interferon-associated genes (Ifi44, Ifi205, STAT2), Nfκb

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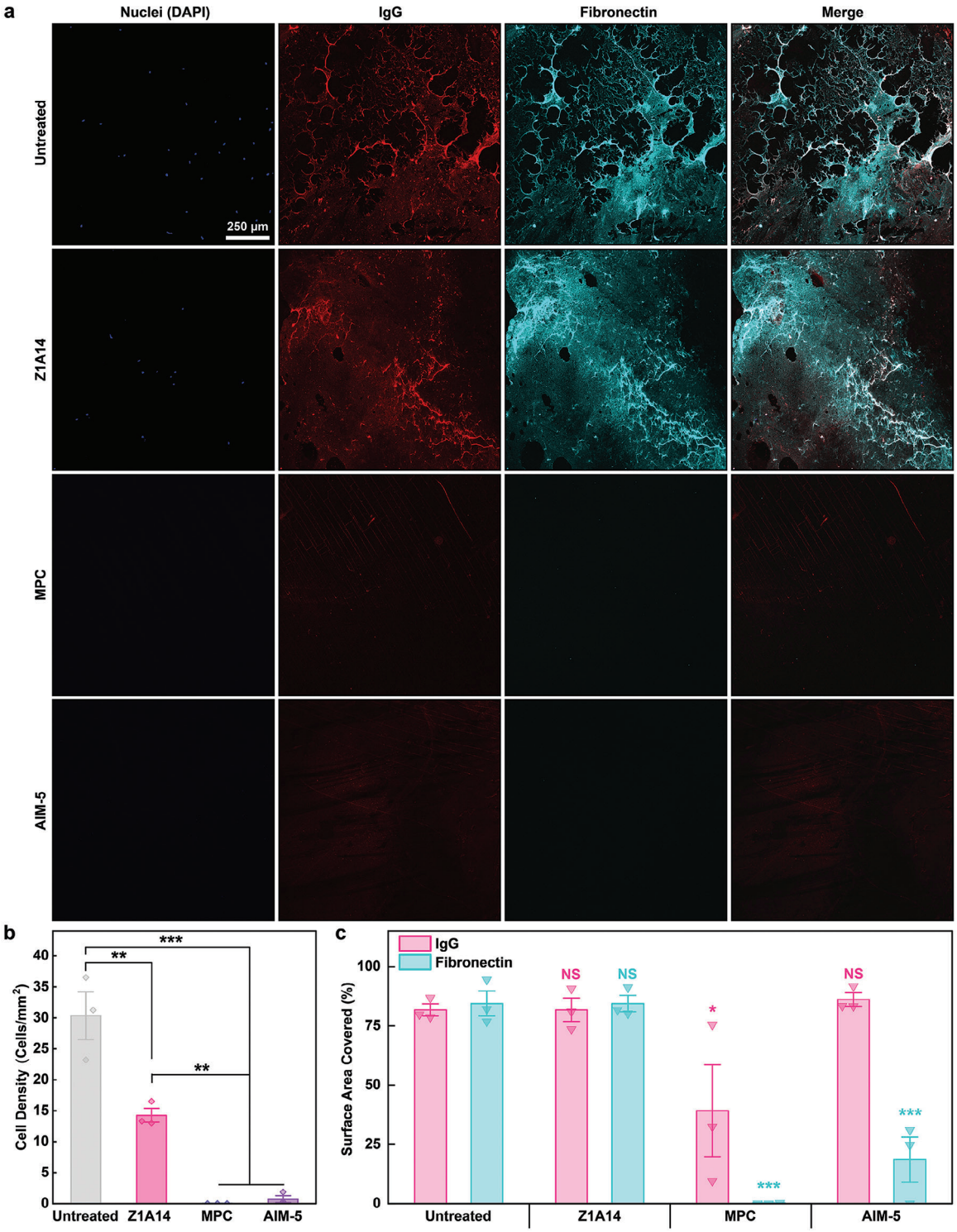


Figure 3. Evaluation of cellular adhesion and protein deposition of lead AIM architecture in vivo. A) Confocal laser scanning micrographs of implant surfaces after 30 min of subcutaneous implantation in C57BL/6 mice ($n = 3$ implants per group). Samples were fixed and stained for cell nuclei using DAPI (blue), IgG proteins (red), and fibronectin (cyan) prior to imaging and image analysis. Note that due to large differences in fluorescent signal across samples, micrographs were not captured under identical settings. B) Plot of adhered cell density for each group. Statistical significance was determined by one-way ANOVA using the Holm-Bonferroni method for multiple comparison correction (**$P < 0.01$ and ***$P < 0.001$). C) Plot of areal coverage by either IgG or fibronectin proteins. Each data point for both cell density and surface area covered represents the average value across at least two different locations per implant (area sampled = 3.23 mm^2). Statistical significance was determined by Dunnett's multiple comparison test relative to the untreated group (*$P < 0.05$ and ***$P < 0.001$).

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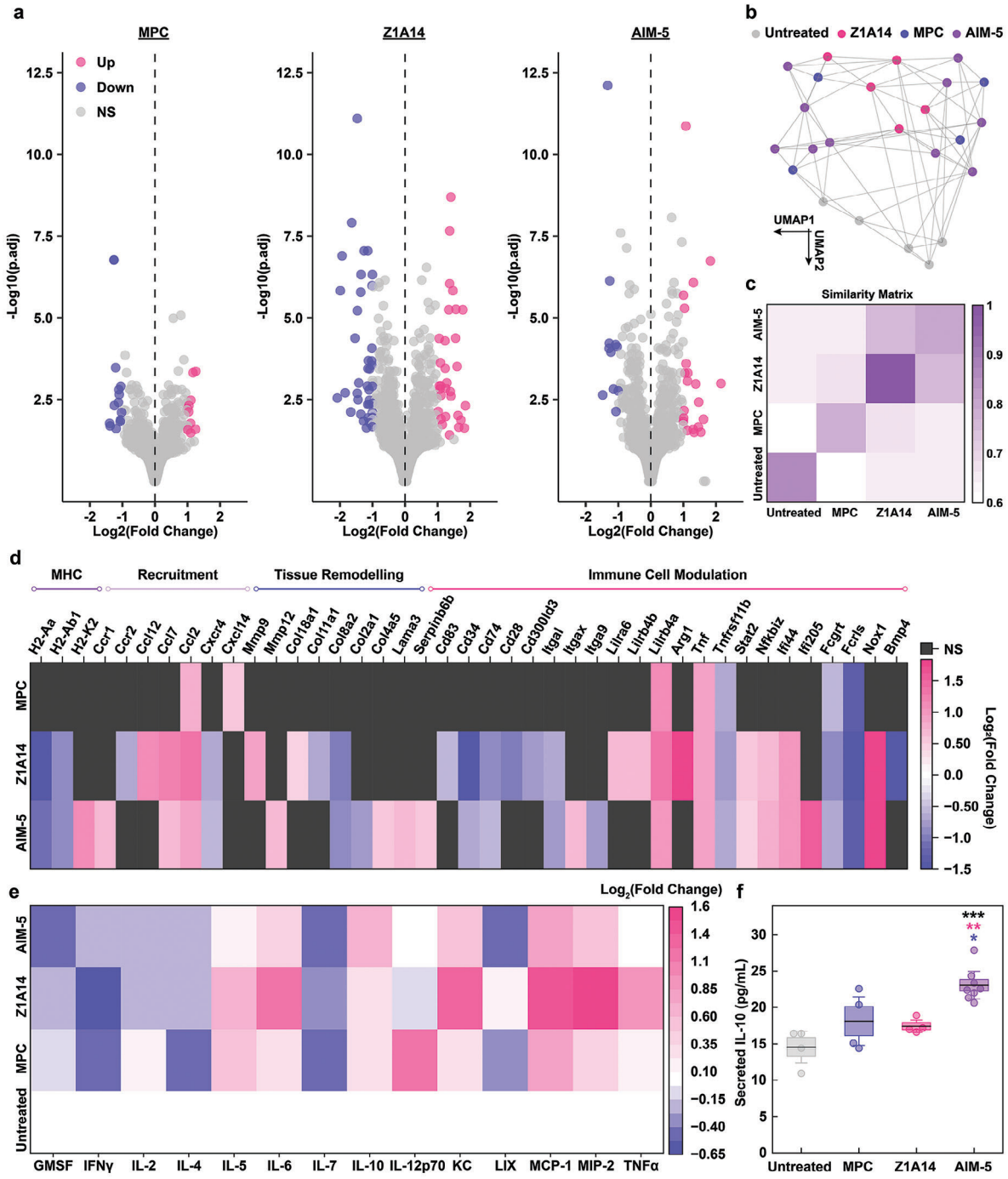


Figure 4. RNA sequencing and secreted cytokine analysis of primary macrophages cultured on antifibrotic surfaces. A) Volcano plots illustrating the average differential gene expression of primary bone marrow derived macrophages cultured on antifibrotic surfaces for 48 h, relative to cells cultured on untreated PDMS. Untreated: $n = 5$, MPC: $n = 4$, Z1A14: $n = 5$, AIM-5: $n = 10$. B) Uniform manifold approximation and projection (UMAP) visualization of the different groups with grey lines connecting each sample to its 5 k-nearest neighbors. C) Plot of the normalized average inverse distance between treatment groups as a marker for similarity. D) Heatmap of expression for selected genes implicated in antigen presentation (MHC: major histocompatibility complex), cellular recruitment, tissue remodeling, and immune cell activation. Grey values indicate that expression was not significantly different from untreated samples. E) Heatmap summary of fold change in secreted cytokine concentration in the cell culture supernatant relative to untreated. Untreated: $n = 4$, MPC: $n = 4$, Z1A14: $n = 4$, AIM-5: $n = 8$. F) Plot of secreted IL-10. Black central lines denote the mean value, boxes indicate standard error, and whiskers represent the 95% confidence interval. Significance was determined by Dunnett's multiple comparison test relative to AIM-5. The color of significance markers indicates comparisons made against untreated (black), MPC (blue), Z1A14 (pink). Significance markers follow: $*P < 0.05$, $**P < 0.01$, $***P < 0.001$.

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inhibitors (Nfkbiz), Fc receptors (Fcgrt, Fcrls), and oxidative stress markers (Nox1, Bmp4). Both materials generally elicited a balanced immune response characterized by a mix of both pro- and anti-inflammatory signatures.

Quantification of secreted cytokines echoed transcriptomic trends (Figure 4E). All materials generally upregulated cellular recruitment factors such as MCP-1, CCL2, and CXCL2, and prompted a mix of pro- and anti-inflammatory phenotypes. IL-1α, IL-1β, IL-13, and IL-17a were nearly absent in all groups, with levels generally below detection limits. Although interleukin expression did not significantly change at the transcript level, significant differences were observed at the protein level. Notably, Z1A14 generated significantly higher levels of IL-6, and AIM-5 stimulated significantly more anti-inflammatory IL-10 (Figure 4F).[81]

These results indicate that despite containing only a small fraction of Z1A14, AIM-5 copolymers still substantially alter the macrophage immune response.

2.6. Lead Architecture Enhances Performance of Multiple Immunomodulatory Moieties in the Subcutaneous Space

We posited that integrating other hydrophobic immunomodulatory moieties with zwitterionic chemistries using the AIM-5 architecture would enhance their performance as well. To test this hypothesis, we synthesized an AIM-5 analog using Z2Y12 (**Figure 5A**), another lead candidate identified in a different library screen from Z1A14.[41] Additionally, we aimed to assess the efficacy of AIM-5 in an alternative implantation site. Therefore, PDMS implants coated with Z1A14 AIM-5 (Z1-AIM) and Z2Y12 AIM-5 (Z2-AIM) were implanted subcutaneously in C57BL/6 mice for 4 weeks (Figure 5B). Figure 5C displays optical micrographs of the explants. Histological evaluation revealed that Z1A14 and Z2Y12 had comparable antifibrotic performance, reducing the average capsule thickness by ≈30% (Figure 5D,E). However, the AIM-5 copolymer forms of both Z1A14 and Z2Y12 significantly outperformed their individual forms, achieving a reduction in fibrotic tissue thickness of over 50% relative to untreated controls. These results suggest that the AIM-5 architecture may broadly improve the antifibrotic properties of hydrophobic immunomodulatory chemistries. Furthermore, the robust performance across different implantation niches, including the peritoneal cavity and subcutaneous space, underscores the versatility of the AIM-5 architecture.

2.7. Lead AIM Architecture Regulates Inflammation in the Fibrotic Tissue Surrounding Implants

To complement analysis of fibrotic capsule thickness, we investigated the inflammatory state of the fibrotic tissue. Experienced pathologists helped analyze H&E samples to identify macrophages by irregularly shaped nuclei and relatively large cytoplasmic volume, neutrophils via multi-lobed nuclei and smaller size relative to the macrophages, lymphocytes by small round nuclei and little visible cytoplasm, and fibroblasts by their elongated spindle shape. In all samples, fibroblasts appear interspersed within the fibrous capsule, and macrophages generally line the implant interface (**Figure 6A**). However, we observed decreased neutrophil and lymphocyte presence in samples treated with immunomodulatory and AIM copolymer coatings, suggesting reduced acute inflammation.

To further characterize the immune response, we performed immunohistochemistry for inducible nitric oxide synthase (iNOS) and arginase 1 (Arg1) as markers of macrophage state (Figure 6B), and alpha smooth muscle actin (α-SMA) as a marker for fibroblast contractile stress (Figure 6C). iNOS is associated with acute inflammation, while Arg1 promotes tissue remodeling and counterbalances iNOS activity by depleting their shared substrate, L-arginine.[82] Excessive iNOS can cause tissue damage and excessive Arg1 activity can lead to pathological fibrosis through overproduction of collagen precursor.[83] Quantification in Figure 6D demonstrates that immunomodulatory and AIM copolymer-treated samples dramatically reduce both iNOS+ and Arg1+ cells relative to untreated and zwitterionic MPC-coated samples. Furthermore, while untreated and MPC surfaces exhibit predominantly iNOS positive cells, indicating sustained inflammation, immunomodulatory and AIM copolymer analogs generally increased the Arg1/iNOS ratio. Notably, AIM copolymers achieve nearly equal ratios of Arg1 and iNOS positive cells. Spatial localization of Arg1 positive cells at implant interfaces suggests that altered cell states result from direct cell-material interactions (Figure 6B). These observations point toward the ability of AIM copolymer surfaces to promote a more balanced macrophage response in a contact-dependent manner, consistent with our in vitro RNA sequencing and cytokine profiling of primary macrophages (Figure 4D–F).

α-SMA expression, indicative of contractile stress exerted by myofibroblasts,[84] was also reduced in immunomodulatory and AIM copolymer samples, which may be benefit soft implants susceptible to mechanical failure.[13,14] These trends in inflammatory phenotype are corroborated by general inflammation scores provided by an expert veterinary pathologist presented with histological sections in a blinded, randomized fashion (Figure 6E). Although the zwitterionic MPC coating yielded a significant reduction in total capsule thickness, the inflammation scores and composition of immune cells within capsules were nearly identical to those of untreated samples. In contrast, immunomodulatory and AIM copolymers appear to achieve their antifibrotic properties through modulation of immune cell population and activity. Combined with results from our fouling studies (Figures 2 and 3), these analyses indicate that the superior antifibrotic performance of AIM copolymers lay in their unique ability to not only reduce initial protein and cellular adsorption but also balance inflammatory phenotypes of adherent cells.

3. Conclusion

Herein we introduce a class of materials endowed with two key properties that inhibit implant fibrosis: fouling resistance and immune regulation.[31–33] Integrating disparate functional monomers, we synthesized a range of antifouling immunomodulatory (AIM) copolymers and evaluated their performance in vivo. Interestingly, we found that random copolymerization of the two functional monomers did not yield a significant change in antifibrotic activity. Our lead polymer, AIM-5, which features a continuous zwitterionic segment coupled with a random copolymer

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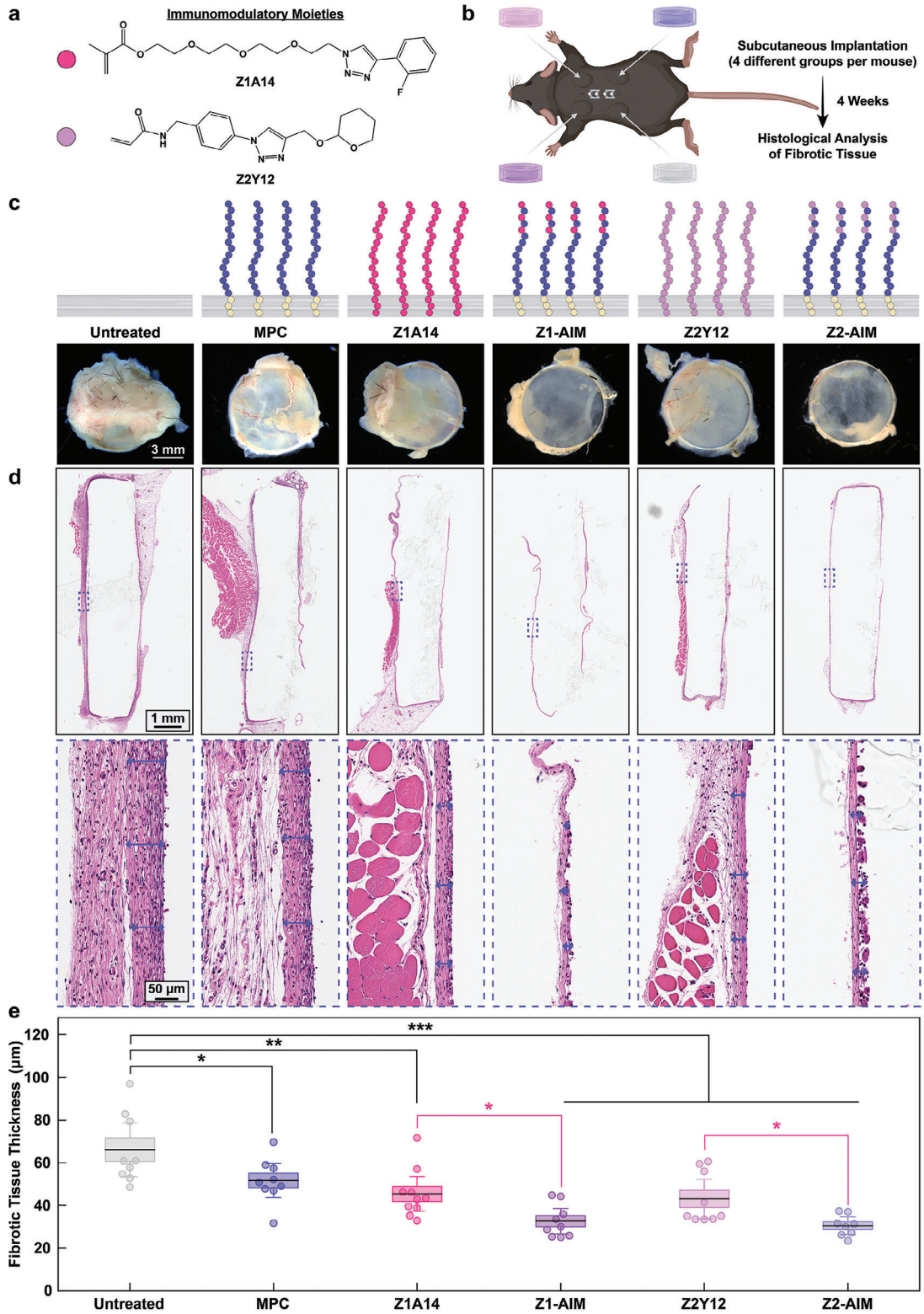


Figure 5. AIM-5 architecture enhances performance of multiple immunomodulatory moieties in the subcutaneous space. A) Chemical structures of two immunomodulatory monomers (Z1A14 and Z2Y12) used to generate different copolymers with the AIM-5 architecture (Z1-AIM and Z2-AIM). These functional groups were identified as lead antifibrotic candidates in two prior library screens by Mukherjee et al.[43] and Vegas et al.[41] B) Schematic overview of in vivo experiment. PDMS implants with 4 different surface chemistries per animal were surgically inserted into the subcutaneous space of immunocompetent mice for 4 weeks prior to explantation and histological evaluation of the fibrotic capsule. C) Ball models of the surface treatments

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tip, significantly outperforms homopolymers of its constituent components in vivo. Investigation into biomolecule affinity, cellular adhesion, and immunological response revealed that this polymer sequence successfully preserves both antifouling and immunomodulatory capabilities. Furthermore, we demonstrated that the AIM-5 architecture achieves superior antifibrotic action in both intraperitoneal and subcutaneous implant sites, as well as when synthesized with two different immunomodulatory monomers. This work establishes copolymer architecture as a critical parameter for achieving synergy between these disparate classes of antifibrotic monomers, enabling AIM copolymers to outperform current best-in-class homopolymers.

We believe these AIM coating architectures may be replicated using these other methods, such as surface-initiated polymerization and click chemistry, in applications where they are more appropriate. The polysilanes coating approach, however, offers interesting possibilities for combination coatings. For example, one might consider creating a cocktail of AIM polysilanes of varying molecular weight, monomer sequence, and density of crosslinkable silane groups to achieve more complex surface topologies, a property that has also shown the ability to enhance the antifouling capabilities of surface-tethered polymer coatings. The ease and versatility of this method contrast with the complex processing required by current micropatterning techniques,[85,86] making AIM polysilane coatings a practical choice for diverse biomedical implants.

Although our current study primarily assessed AIM coatings on PDMS substrates, efficient grafting and consistent changes in physicochemical properties achieved on both ceramic and metallic substrates supports broader application (Table S3 and Figure S10, Supporting Information). However, further in vivo testing is necessary to verify whether enhanced antifibrotic properties hold across other types of implants, implant niches, and animal species. Likewise, our initial characterization of adsorption profiles and immune response phenotype lays the foundation for more detailed studies required to elucidate the underlying mechanisms of action. Nevertheless, the insights gained from our studies on copolymer architecture engender a new degree of freedom in the design of antifibrotic biomaterials by motivating future investigations into more advanced architectures and new combinations of different functional chemistries.

4. Experimental Section

Synthesis of Immunomodulatory Monomers: Unless otherwise noted, all manipulations were carried out under an Argon atmosphere using standard Schlenk-line techniques. All glassware was oven-dried at 150 °C overnight before use. Organic solvents including anhydrous tetrahydrofuran (THF), dichloromethane (DCM), dimethylformamide (DMF) were purchased from Sigma–Aldrich and used as received under Argon atmosphere. Z2Y12 amine was synthesized following a previously reported procedure.[41] All other starting materials were obtained from commercial sources or outsourced and were used as received. Nuclear magnetic resonance (NMR) spectra were obtained on a Bruker Advance 500 spectrometer (500.18 MHz). Deuterated solvents were purchased from Cambridge Isotope Laboratories or Sigma–Aldrich. Chromatography was performed using silica gel on an automated flash chromatography instrument (Teledyne ISCO, CombiFlash) using RediSep Gold Flash Cartridges. NMR spectra were reported as parts per million (multiplicity, coupling constant in Hz, # of protons). Multiplicity was described as singlet (s), doublet (d), doublet of doublet (dd), doublet of triplet (dt), doublet of doublet of doublet (ddd), triplet, (t), quartet (q), and multiplet (m).

Synthesis of Z1A14 Intermediate-1: To synthesize Z1A14 monomer, an alcohol terminated intermediate was first synthesized. Copper(I)Iodide (1.4 g, 7.3 mmol, 0.5 eq.) was suspended in 20 mL of anhydrous methanol in a 100 mL round-bottom flask equipped with a magnetic stir bar. 1-Ethynyl-2-fluorobenzene (1.9 g, 16.1 mmol, 1.1 eq.), N,N-Diisopropylethylamine (7.6 mL, 43.8 mmol, 3.0 eq.) was then added and stirred for 5 min. Later, 2-(2-(2-(2-azidoethoxy)ethoxy)ethoxy)ethan-1-ol (3.2 g, 14.6 mmol, 1.0 eq.) was slowly added dropwise at 0 °C and stirred overnight. The following day, the solution was filtered through celite, and the solvent was removed *in vacuo*. Products were purified via silica gel flash column chromatography using DCM and methanol as eluent from 0 to 10% for 60 min, and dried by rotary evaporation to afford Intermediate-1 (2-(2-(2-(2-(4-(2-fluorophenyl)-1*H*-1,2,3-triazol-1-yl)ethoxy)ethoxy)ethoxy)ethan-1-ol) as a colorless liquid; Yield: 70.8%, 3.51 g.

^{1}H NMR (500 MHz, $CDCl_3$) δ 8.33 (td, J = 7.7, 1.9 Hz, 1H), 8.18 (d, J = 3.7 Hz, 1H), 7.37 – 7.23 (m, 2H), 7.16 (ddd, J = 11.1, 8.1, 1.3 Hz, 1H), 4.65 (t, J = 5.1 Hz, 2H), 3.95 (t, J = 5.0 Hz, 2H), 3.80 – 3.68 (m, 2H), 3.70 – 3.46 (m, 10H).

^{13}C NMR (126 MHz, $CDCl_3$) δ 160.19, 158.22, 140.93, 129.3 (d, J = 30 Hz), 127.86 (d, J = 15 Hz), 124.6 (d, J = 10 Hz), 124.2 (d, J = 50 Hz), 118.7 (d, J = 50 Hz), 115.6 (d, J = 85 Hz), 72.4, 70.7, 70.6, 70.5, 70.3, 69.6, 61.6, 54.6, 50.5, 42.8.

^{19}F NMR (471 MHz, $CDCl_3$) δ -114.8.

Synthesis of Z1A14 Monomer: Z1A14 monomer was obtained by esterifying the alcohol-terminated intermediate to generate a polymerizable methacrylate handle. The reaction began by charging an oven-dried 250-mL round-bottom flask with a magnetic stir bar and Intermediate-1 (2-(2-(2-(2-(4-(2-fluorophenyl)-1*H*-1,2,3-triazol-1-yl)ethoxy)ethoxy)ethoxy)ethan-1-ol) (3.51 g, 10.3 mmol,) and then adding 100 mL of anhydrous THF. After dissolving Intermediate-1, triethylamine (3.63 mL, 25.9 mmol, 2.5 eq.) was added and stirred for 5 min. Later, methacryloyl chloride (2.53 mL, 25.9 mmol, 2.5 eq.) was slowly added dropwise with the flask held on ice and allowed to stir for two days. After the reaction, THF was removed *in vacuo*. Products were purified via silica gel flash column chromatography using DCM and methanol as eluent from 0 to 10% for 60 min, and dried by rotary evaporation to afford Z1A14 monomer as a pale yellow, nearly colorless liquid; Yield: 31.3%, 1.32 g.

^{1}H NMR (500 MHz, $CDCl_3$) δ 8.32 (td, J = 7.6, 1.9 Hz, 1H), 8.13 (d, J = 3.8 Hz, 1H), 7.36 – 7.24 (m, 2H), 7.15 (ddd, J = 11.1, 8.1, 1.4 Hz, 1H), 6.15 – 6.11 (m, 1H), 5.57 (p, J = 1.6 Hz, 1H), 4.63 (t, J = 5.1 Hz, 2H), 4.32 – 4.26 (m, 2H), 3.98 – 3.91 (m, 2H), 3.75 – 3.69 (m, 2H), 3.64 (d, J = 4.1 Hz, 8H), 1.95 (t, J = 1.3 Hz, 3H).

^{13}C NMR (126 MHz, $CDCl_3$) δ 167.4, 158.2, 141.01(d, J = 10 Hz), 136.2, 129.28(d, J = 35 Hz), 127.88(d, J = 15 Hz), 125.7, 124.63 (d, J = 10 Hz),

tested (top) and representative micrographs of explanted samples (bottom). D) Histological cross sections of explants stained with hematoxylin and eosin (H&E) revealing fibrotic tissue characterized by dense, striated morphology. Blue arrows delineate the fibrotic tissue thickness and blue dashed borders indicate the location of higher magnification images presented below. E) Quantification of the average thickness of fibrotic capsules around implants. Untreated: n = 9, MPC: n = 9, Z1A14: n = 10, Z1-AIM: n = 9, Z2Y12: n = 9, Z2-AIM: n = 8. Each data point represents the average of at least 15 discrete measurements taken around the entire circumference of each sample. Black central lines denote the mean value, boxes indicate standard error, and whiskers represent the 95% confidence interval. Black brackets denote statistical comparisons made against the untreated group using Dunnet's multiple comparison test, while pink brackets denote unpaired, two-tailed, t-tests comparing previously reported immunomodulatory chemistries and their respective AIM-5 analog. Significance markers follow: *P < 0.05, **P < 0.01, ***P < 0.001.

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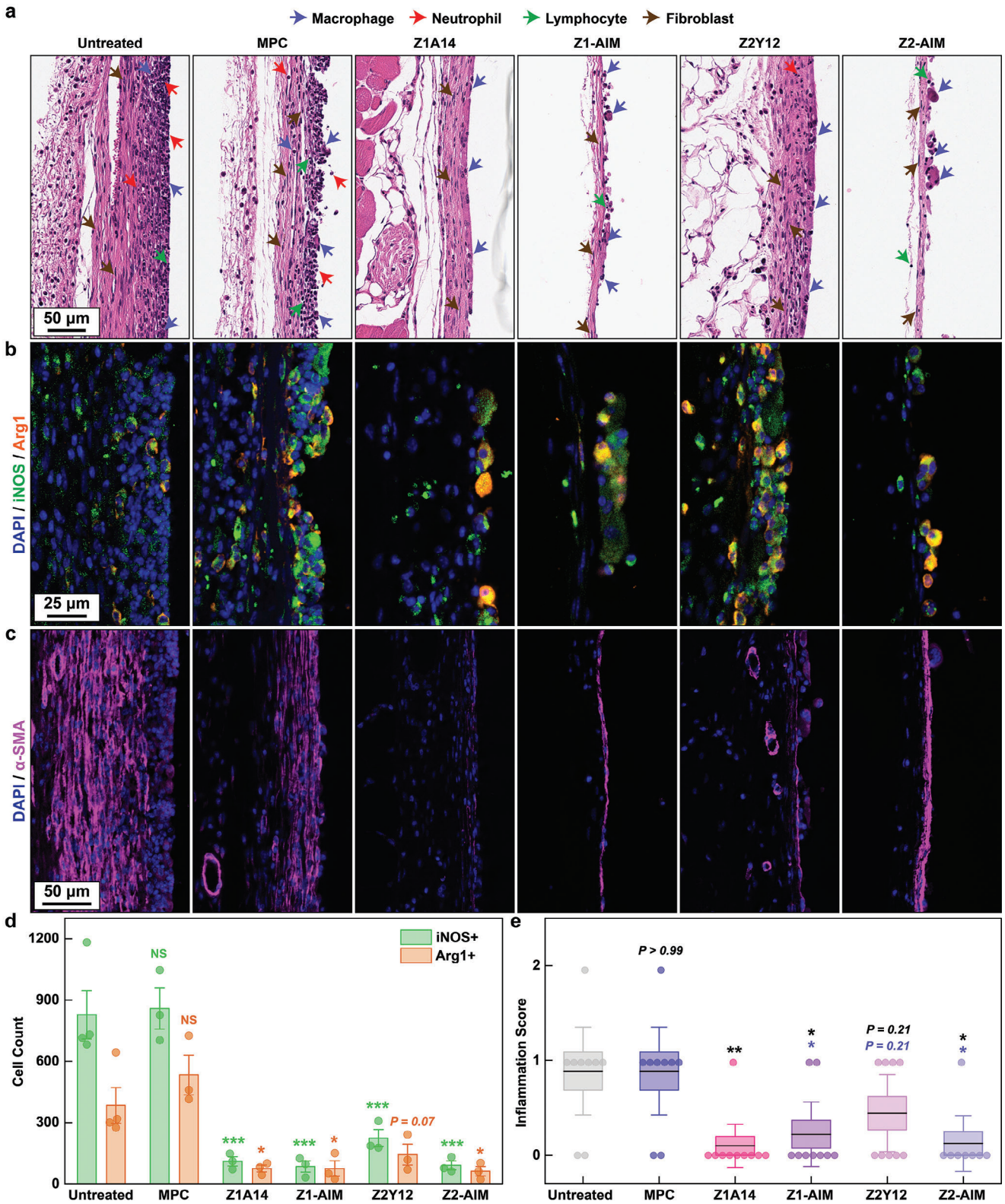


Figure 6. AIM-5 analogs of immunomodulatory coatings balance the inflammatory phenotype in fibrotic tissue surrounding subcutaneous implants. A) H&E-stained sections from subcutaneous implants retrieved after 4 weeks, annotated to denote cell types relevant to inflammation. The implant interface is located on the right-hand side for all histology and immunohistochemistry images. B) Sections stained by fluorescent immunohistochemistry for nuclei (DAPI, blue), inducible nitric oxide synthase (iNOS, green), and arginase 1 (Arg1, orange). C) Fluorescently stained sections for nuclei (DAPI,

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124.09(d, J = 50 Hz), 118.72(d, J = 55 Hz), 115.73(d, J = 90 Hz), 70.77, 70.75, 70.65, 70.58, 69.5, 69.1, 63.8, 50.4, 18.3.

^{19}F NMR (471 MHz, CD_2Cl_2) δ -114.8

Synthesis of Z2Y12 Monomer: Z2Y12 monomer resulted from an amidation reaction between Z2Y12 amine and an acyl chloride. The reaction involved charging an oven-dried 500 mL round-bottom flask with a magnetic stir bar, Z2Y12 amine ((4-(4-(((tetrahydro-2*H*-pyran-2-yl)oxy)methyl)-1*H*-1,2,3-triazol-1-yl)phenyl)methanamine) (6.0 g, 20.8 mmol,), and 200 mL of anhydrous THF. N,N-Diisopropylethylamine (5.53 mL, 31.2 mmol, 1.5 eq.) was then added and stirred for 5 min. Later, acryloyl chloride (2.55 mL, 31.2 mmol, 1.5 eq.) was slowly added dropwise with the flask held on ice and then allowed to stir overnight. The next day, THF was removed *in vacuo*. Products were purified via silica gel flash column chromatography using DCM and methanol as eluent from 0 to 10% for 60 min, and dried by rotary evaporation to afford Z2Y12 monomer as a pale yellow solid; Yield: 71.6%, 5.10 g.

^{1}H NMR (500 MHz, $CDCl_3$) δ 7.98 (s, 1H), 7.68 (d, J = 8.5 Hz, 2H), 7.45 (d, J = 8.5 Hz, 2H), 6.37 (dd, J = 17.0, 1.4 Hz, 1H), 6.30 (s, 1H), 6.20 (dd, J = 16.9, 10.3 Hz, 1H), 5.72 (dd, J = 10.3, 1.5 Hz, 1H), 4.97 (d, J = 12.4 Hz, 1H), 4.85 – 4.78 (m, 1H), 4.74 (d, J = 12.4 Hz, 1H), 4.60 (d, J = 6.0 Hz, 2H), 3.96 (ddd, J = 11.1, 8.0, 3.0 Hz, 1H), 3.66 – 3.55 (m, 1H), 1.95 – 1.71 (m, 3H), 1.71 – 1.51 (m, 4H).

^{13}C NMR (126 MHz, $CDCl_3$): δ 165.60, 146.00, 139.13, 136.28, 130.47, 129.10, 127.23, 120.78, 120.77, 98.45, 62.47, 60.55, 42.94, 30.49, 25.36, 19.43.

Synthesis of Zwitterionic MPC Polymers with Different Grafting Modes: We synthesized a range of zwitterionic polymers of 2-methacryloyloxyethyl phosphorylcholine (MPC) (Sigma Aldrich, 730 114) using photo-induced atom transfer radical polymerization (photo-ATRP) to assess different grafting chemistries (azide-alkyne click chemistry, triblock silanes, diblock silanes) affect the fibrotic response. The protocol for surface-initiated polymerization of MPC by traditional ATRP appears in the later section describing surface functionalization of silicone implants. All methods had comparable performance, so diblock silanes were used for most of the presented studies since they featured slightly higher performance and do not require pre-functionalization of the PDMS implants with other chemical moieties.

PhotoATRP utilized a ligand complex formed from $CuBr_2$ and tris(2-pyridylmethyl)amine (TPMA) (Sigma–Aldrich, 723 134). MPC polymers designated for either click chemistry grafting or triblock silane grafting utilized ethylene bis(2-bromoisobutyrate) (Sigma–Aldrich, 723 177) as a bifunctional ATRP initiator, while polymers designated for diblock silane polymers utilized ethyl α-bromoisobutyrate (EBiB) (Sigma–Aldrich, E14403) as a monofunctional ATRP initiator. The 3-(Trimethoxysilyl)propyl methacrylate (TESPMA) monomer (Gelest, SIM6487.3) was filtered through a column of activated alumina before use to remove polymerization inhibitor. All reactions were carried out under vigorous stirring with ultraviolet (UV) irradiation mediated by a 15-Watt, 365 nm UV lamp (Analytik Jena, 95-0042-06) with black light blue (BLB) bulbs positioned 12 cm away from the reaction vessel. NMR analyses were performed in deuterated methanol (Sigma–Aldrich, 422 878). Polymerization was monitored by taking aliquots of the degassed reaction mixture before, during, and after UV irradiation and quantifying the disappearance of the MPC vinyl peak (δ = 5.50 ppm) or TESPMA vinyl peak (δ = 5.42 ppm) using the DMF formyl peak (δ = 7.84 ppm) as a reference at each time point.

PhotoATRP of Bifunctional MPC Macroinitiators: A typical photoATRP reaction using a bifunctional initiator involved dissolving MPC (2657.4 mg, 9 mmol, 80 eq.) in 8.865 mL of methanol in a 20 mL glass vial with a magnetic stir bar. After dissolving the monomer, ethylene bis(2-bromoisobutyrate) (40.5 mg, 112.5 µmol, 1 eq.) was added. A ligand stock with 10 mM $CuBr_2$ and 60 mM TPMA was prepared in methanol and 1.125 mL of this stock was added to the reaction mix to administer $CuBr_2$ (2.51 mg, 11.25 µmol, 0.1 eq.) and TPMA (19.6 mg, 67.5 µmol, 0.6 eq.). 100 µL of DMF was added as an internal standard for NMR analysis and an additional 910 µL of methanol was added to account for evaporation during degassing. The reaction mix was sealed with a rubber septum and electrical tape before sparging with argon for 25 min to remove dissolved oxygen. Overnight UV irradiation led to monomer conversions typically exceeding 90%. Purification involved dialyzing the mixture against deionized water followed by lyophilization to obtain bifunctional MPC macroinitiators.

Azidation of Bifunctional MPC Polymers: Several methods were tested for installing azido functional groups on the ends of MPC polymers for the purpose of grafting by copper-catalyzed azide-alkyne cycloaddition (CuAAC) click reaction to the surface of alkyne-functionalized implants. Initially end group transformation utilized sodium azide (NaN_3) with heating or by copper assisted azidation. However, the best grafting results were achieved via additional polymer blocks generated by first extending the bifunctional MPC polymer with glycidyl methacrylate (GMA) (Sigma–Aldrich, 779 342), and then utilizing a ring opening reaction between NaN_3 and the epoxide side groups to install multiple azido functional groups.

GMA photoATRP and subsequent azide-epoxide ring opening took place in a one pot scheme immediately following the synthesis of bifunctional MPC macroinitiators, without purification. After the MPC monomer conversion exceeded 90%, 10 eq. of GMA was added to the reaction mixture, accounting for mass loss from prior NMR samples, along with 1 mL of methanol to account for future bubbling evaporation. The reaction mix was again degassed by argon bubbling for 25 min and irradiated with UV light overnight. NMR analysis verified quantitative conversion of the monomer before the addition of NaN_3 (25 eq.) and NH_4Cl (25 eq.) and heating at 60 °C for 48 h. After dialysis against deionized water and lyophilization, NMR analysis of the purified product revealed complete extinction of the epoxide protons (δ = 2.85 ppm and 2.7 ppm), indicating successful ring opening.

Silane Extension to Generate Triblock MPC Silane Polymers: Purified bifunctional MPC macroinitiator (177.4 mg, 8 µmol, 1 eq.) was redissolved in 3.807 mL of ethanol in a 20 mL glass vial. Next, alumina-filtered TESPMA (34.9 mg, 120 µmol, 15 eq.) was pipetted in. A ligand stock with 10 mM $CuBr_2$ and 60 mM TPMA was prepared in ethanol and 80 µL of this stock was added to the reaction mix to administer $CuBr_2$ (0.178 mg, 0.8 µmol, 0.1 eq.) and TPMA (1.39 mg, 4.8 µmol, 0.6 eq.). 80 µL of DMF was added as an internal standard for NMR analysis and an additional 1 mL of additional ethanol was added to account for evaporation during degassing. The reaction mix was sealed with a rubber septum and electrical tape before sparging with argon for 30 min to remove dissolved oxygen. Overnight UV irradiation achieved a monomer conversion of 72.3%. To purify the product, the reaction mix was concentrated by rotary evaporation until viscous before precipitation in 80:20 Acetone:THF. The precipitates were rinsed three times with the precipitating solvent before drying under vacuum. The white solids were immediately dissolved in ethanol to a concentration of 10 mg mL^{-1}, filtered through a 450 nm syringe filter and stored under argon at 4 °C.

Synthesis of diblock MPC Silane Polymers: The synthesis of diblock silanes utilized as MPC coatings in the majority of this studies employed a monofunctional ATRP initiator. MPC (2.126 g, 7.2 mmol, 80 eq.) and 6.99 mL of methanol were combined in a 20 mL glass vial with a magnetic stir bar. After dissolving the monomer, EBiB (17.6 mg, 90 µmol, 1

blue) and alpha smooth muscle actin (α-SMA, magenta). D) Quantification of cells stained positive for iNOS and Arg1. Each data point represents the total cell count across the long edge of an explanted sample. Control: n = 4, all other groups: n = 3. Error bars represent standard error. Significance markers indicate comparisons against untreated for either iNOS+ (green) or Arg1+ (orange) using Dunnett's multiple comparisons test. E) Degree of inflammation assessed by a veterinary pathologist in a randomized, blinded manner (0 = normal, 1 = mild, 2 = moderate). Untreated: n = 9, MPC: n = 9, Z1A14: n = 10, Z1-AIM: n = 9, Z2Y12: n = 9, Z2-AIM: n = 8. Black central lines denote the mean value, boxes indicate standard error, and whiskers represent the 95% confidence interval. Statistical significance was determined from Dunnett's multiple comparison test. The color of the significance markers indicates comparisons made relative to untreated (black) or MPC (blue). Significance markers follow: *P < 0.05, **P < 0.01, ***P < 0.001.

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eq.) was pipetted in. A ligand stock containing 10 mM $CuBr_2$ and 60 mM TPMA was prepared in methanol and 0.9 mL of this stock was added to administer $CuBr_2$ (2 mg, 9 µmol, 0.1 eq.) and TPMA (19.6 mg, 54 µmol, 0.6 eq.). 100 µL of DMF was added as an internal standard for NMR analysis and an additional 1 mL of methanol was added to account for evaporation during degassing. The vial was sealed with a rubber septum and electrical tape before sparging with argon for 25 min to remove dissolved oxygen. Overnight UV irradiation with stirring resulted in 97.9% monomer conversion. Dialyzing the mixture against deionized water followed by lyophilization afforded the MPC macroinitiator.

In a typical silane chain extension reaction, purified MPC macroinitiator (93.3 mg, 4 µmol, 1 eq.) was dissolved in 1.897 mL of ethanol in an 8 mL glass vial equipped with a magnetic stir bar. Next, alumina-filtered TESPMA (23.2 mg, 80 µmol, 20 eq.) was added using a micropipette. 40 µL of the ethanol-based copper ligand stock was added to deliver $CuBr_2$ (8.92 µg, 0.4 µmol, 0.1 eq.) and TPMA (0.697 mg, 2.4 µmol, 0.6 eq.). 40 µL of DMF was included as an internal standard, and an additional 1 mL of ethanol compensated for evaporation during degassing. The vial was sealed with a rubber septum and electrical tape before sparging with argon for 25 min to remove dissolved oxygen. Overnight UV irradiation with stirring resulted in 67% monomer conversion. The reaction mix was concentrated by rotary evaporation until a viscous liquid was obtained before precipitating in 80:20 Acetone:THF. The precipitates were rinsed three times with the precipitating solvent before drying under vacuum. The dried diblock copolymer was immediately dissolved in ethanol to a concentration of 10 mg mL^{-1}, filtered through a 450 nm syringe filter, and stored under argon at 4 °C.

Synthesis AIM Copolymers: AIM copolymers were synthesized by photoATRP using a similar macroinitiator-based scheme as described above. However, degassing was carried out by a minimum of 5 freeze-pump-thaw cycles in an oven-dried 10 mL Schlenk flask as we found this method yielded more efficient polymerization, particularly for steps involving immunomodulatory monomers. A digital vacuum gauge (CPS, VG200) installed on the Schlenk line ensured the pumping pressure remained below 6 mbar. The TESPMA monomer was filtered through a column of activated alumina before use to remove polymerization inhibitor. An ethanol-based ligand stock containing 10 mM $CuBr_2$ and 60 mM TPMA was used for all steps. All reactions were carried out with vigorous stirring under positive argon pressure. UV irradiation was mediated by a 15-Watt, 365 nm UV lamp (Analytik Jena, 95-0042-06) with black light blue (BLB) bulbs positioned 12 cm away from the reaction vessel. NMR analyses were performed in deuterated methanol unless otherwise stated. Polymerization was monitored by taking aliquots of the degassed reaction mixture before, during, and after UV irradiation and quantifying the disappearance of the monomer vinyl peaks of Z1A14 (δ = 5.60 ppm), Z2Y12 (δ = 5.72 ppm), MPC (δ = 5.64 ppm), or TESPMA vinyl peak (δ = 5.42 ppm) using either the DMF formyl peak (δ = 7.84 ppm) or a Z1A14 aromatic peak (δ = 8.13 ppm) as a reference at each time point. A summary of all the monomer ratios, reaction steps, and corresponding results for all AIM copolymers appears in Table S1 (Supporting Information).

Random Copolymerization of Z1A14 and MPC: Synthesis of AIM-1, AIM-2, AIM-3, and AIM-5 began by synthesizing a random copolymer of Z1A14 and MPC to form the immunomodulatory tip. A typical reaction mixture contained 10 mM EBiB (1 eq.) and the corresponding amount of Z1A14 (AIM-1: 25 eq., AIM-2: 65 eq., AIM-3: 25 eq., AIM-5: 25 eq.) and MPC (AIM-1: 25 eq., AIM-2: 65 eq., AIM-3: 105 eq., AIM-5 25 eq.) prepared in 1:1 mixture of ethanol and chloroform. The copper ligand stock was then added to deliver 0.1 eq. of $CuBr_2$ and 0.6 eq. of TPMA. Reaction volumes typically ranged from 1 – 2 mL. Freeze-pump-thaw degassing and UV irradiation overnight completed the reaction. Products were dialyzed against 20% ethanol in water for 12 h before further dialysis in deionized water. The full contents of the dialysis cassette, which in some cases included plasticized precipitates, were transferred to a flask for lyophilization. The obtained solids were then stored at 4 °C under argon until used.

Homopolymerization of Z1A14: Synthesis of AIM-4 began by polymerizing Z1A14. EBiB (3.6 mg, 10 µmol, 1 eq.), Z1A14 (101.9 mg, 250 µmol, 25 eq.), and 10 µL of DMF, were dissolved in 774 µL of 1:1 mixture of ethanol and chloroform in a Schlenk flask. 100 µL of the ligand stock contributed $CuBr_2$ (0.223 mg, 1 µmol, 0.1 eq.) and TPMA (0.174 mg, 0.6 µmol, 0.6 eq.). Degassing by freeze-pump-thaw and overnight UV irradiation facilitated photoATRP.

Chain Extension by MPC: AIM-4 and AIM-5 involved chain extension by MPC using a macroinitiator comprising either a homopolymer of Z1A14 or a copolymer of Z1A14 and MPC, respectively. Chain extension of Z1A14 homopolymer for AIM-4 was directly carried out in a one-pot fashion. Before adding MPC, ethanol was added to achieve a solvent ratio of 2:1 ethanol to chloroform. For AIM-5, lyophilized copolymer was redissolved in 3:1 mixture of ethanol and chloroform to a final concentration of 5 mM (1 eq.) along with 1% v/v DMF and the appropriate amount of ligand stock to incorporate 0.1 eq. of $CuBr_2$ and 0.6 eq. of TPMA. In both cases, 80 eq. of MPC was added to the reaction mixture. Reaction volumes typically ranged from 1 – 2 mL. After dissolution, deoxygenation by freeze-pump-thaw and overnight UV irradiation facilitated polymerization. The reaction mix was concentrated by rotary evaporation until viscous before precipitation in 80:20 Acetone:THF. Precipitates were rinsed with the precipitating solvent and redissolved in ethanol before an additional round of precipitation. Purified products were then dried under vacuum and stored at 4 °C under argon until used.

Chain Extension by TESPMA: AIM polymers were completed by adding a block of TESPMA to enable grafting to implant surfaces. Corresponding macroinitiators were dissolved in ethanol to a final concentration of 4 mM (1 eq.) along with 1% v/v DMF and appropriate amount of ligand stock to deliver 0.1 eq. of $CuBr_2$ and 0.6 eq. of TPMA. Next, alumina-filtered TESPMA (AIM-1: 10 eq., all other AIM polymers: 18 eq.) was added. Freeze-pump-thaw and overnight UV irradiation facilitated polymerization. The reaction mix was concentrated by rotary evaporation until viscous before precipitation in 80:20 Acetone:THF. Precipitates were rinsed with the precipitating solvent and redissolved in ethanol before an additional round of precipitation. Purified products were then dried under vacuum and immediately dissolved in ethanol to a concentration of 10 mg mL^{-1}, filtered through a 450 nm syringe filter, and stored under argon at 4 °C.

Characterization of Polysilane Stability and Size Distribution in solution: Ethanolic solutions of polysilane copolymers (10 mg mL^{-1}) were evaluated by dynamic light scattering (Zetasizer Nano, Malvern) at 25 °C with 3 discrete scans per sample. Intensity size distributions are reported along with polydispersity indices. Samples were measured immediately after purification, as well as after several months of storage at 4 °C under argon.

Preparation of Silicone Implants: Polydimethylsiloxane (PDMS) was prepared using a Sylgard 184 kit (Dow) by mixing the elastomer base with the curing agent at a 10:1 ratio and pouring into a plastic petri dish. The PDMS mixture was degassed in a vacuum chamber for at least 30 min until all visible bubbles disappeared and then cured at 60 °C overnight. After relieving the solid PDMS elastomer from the petri dish, 8 mm (Integra) or 1 cm (Acuderm) biopsy punches were used to obtain cylindrical samples (Thickness = 1.5 mm). Using fresh biopsy punches minimized roughness along the cut edges for samples designated for implantation.

Surface Functionalization of Silicone Implants: *Polysilane grafting*: To graft PDMS with silane containing polymers, the PDMS disks were treated with air plasma (Harris Plasma Chamber, high radio frequency) with pressure between 500 and 700 mtorr for 90 s on each side. Plasma treated samples were immediately dropped into a 5 mg mL^{-1} of the given silane polymer solution in ethanol with 5% v/v deionized water and left on a plate shaker at 300 rpm overnight, ensuring that samples did not overlap. Grafted samples were then washed three times with ethanol and dried in an oven at 75 °C for 45 min.

Plasma-induced surface polymerization: Samples were functionalized with immunomodulatory Z1A14 and Z2Y12 by plasma-induced surface polymerization following previous reports.[43,53] PDMS disks were air plasma treated on both sides and dropped into a solution of 0.02 M solution of either Z1A14 or Z2Y12 monomer in 5% v/v dimethyl sulfoxide in toluene and agitated on a plate shaker at 300 rpm for 2 h. Grafted samples were then washed three times each in methanol, ethanol, and water and a final time with ethanol. Samples were then dried overnight in a vacuum desiccator.

Surface-initiated ATRP: A "graft-from" polymerization scheme for MPC using traditional atom transfer radical polymerization (ATRP) was

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developed following previous reports.[6] PDMS disks were first grafted with an ATRP initiator by treating with air plasma on both sides for 90 s and incubating in a silane bath with 95% v/v ethanol, 3% v/v water, and 2% v/v (3-trimethoxysilyl)propyl 2-bromo-2-methylpropionate overnight at room temperature on a plate shaker at 300 rpm. Samples were then washed three times with ethanol and dried at 60 °C. The polymerization reaction involved charging a 10 mL Schlenk flask with a magnetic stir bar, MPC monomer (833 mg, 2.82 mmol), and 5 mL of a 1:1 mixture of methanol and deionized water. After thoroughly dissolving the monomer, a maximum of 3 PDMS samples functionalized with the ATRP initiator were loaded into the flask before sealing with a rubber septum and placing on an ice bath. Sparging with nitrogen for 20 min removed dissolved oxygen. Removing the ice bath and adding CuBr (4.04 mg, 0.028 mmol) and 2,2′-bipyridine (bpy) (8.81 mg, 0.056 mmol) under strong nitrogen flow initiated the reaction. The reaction ran for 3 h at room temperature with stirring under continuous weak nitrogen flow. Grafted samples were removed and washed 3 times with methanol, thrice with 10x DPBS overnight, and again with deionized water. Any samples that were stuck together were discarded to ensure proper surface functionalization.

"Click" chemistry grafting: Copper-catalyzed azide-alkyne cycloaddition (CuAAC) was used for click chemistry-based attachment of MPC polymers functionalized with polymeric azido blocks. PDMS samples were first functionalized with linear alkynes by treating with air plasma on both sides for 90 s and incubating in a silane bath with 95% v/v ethanol, 3% v/v water, and 2% v/v O-(propargyl)-N-(triethoxysilylpropyl) carbamate overnight at room temperature on a plate shaker at 300 rpm. Samples were then washed three times with ethanol and dried at 60 °C. The CuAAC reaction involved briefly degassing a methanol solution containing the azide-functionalized MPC polymer (0.02 mM), $CuBr_2$ (1 mM), tris-hydroxypropyltriazolylmethylamine (1.1 mM), and ascorbate (5 mM) by sparging with argon for 1 min. Alkyne-functionalized PDMS samples were then immersed in the solution, ensuring samples did not overlap, and the reaction vessel was purged with argon. The reaction ran overnight before washing 3 times with methanol and drying at 60 °C.

Sample Storage and Preparation for Implantation: Regardless of surface treatment, samples designated for surface characterization were kept dry in a desiccator. Samples designated for implantation or cell culture were further washed using 70% ethanol in a sterile biosafety cabinet, twice again with sterile water, and then stored in sterile water at 4 °C until use.

Polysilane Grafting on Ceramic and Metallic Substrates: Glass microscope slides were obtained from VWR (catalog no. 16004–422) and titanium aluminum vanadium alloy foil (thickness = 50 µm) was purchased from Goodfellow (TI01-FL-000151). Samples were cut into centimeter-scale coupons and sonicated in ethanol for 30 s before drying in a 60 °C oven. Cleaned samples were then plasma treated, reacted, washed, and dried following the same procedure described above for PDMS implants.

Water Contact Angle Measurement: Samples mounted on a stage with a white, diffuse backlight received a 5-µL drop of deionized water. A smartphone camera outfitted with a macro lens attachment (Xenvo) then captured sessile drop images before digital contact angle analysis using ImageJ.

X-Ray Photoelectron Spectroscopy: XPS measurements were performed on a PHI Versaprobe II X-ray Photoelectron Spectrometer at a pressure below 2×10^{-8} Torr. Dual beam charge neutralization was used for insulating samples. Data was acquired and analyzed using SmartSoft (PHI) software to quantify elemental composition. Z-height was individually optimized for each measurement using the default automated protocol. The acquired spectra were processed using a Savitzky-Golay filter to improve the signal-to-noise.

Scanning Electron Microscopy: Samples were mounted on aluminum specimen stubs using double-sided carbon tape. Sputter coating (Denton) with platinum (Thickness <15 nm) imparted conductive surfaces. Application of silver conductive paint (Electron Microscopy Sciences) along the side edges of samples ensured robust electrical connection between the top surface and the underlying carbon tape. Micrographs were recorded on a Zeiss Gemini 450 field emission scanning electron microscope with voltage set to 1.5 keV and current set to 500 pA.

Surgical Implantation and Retrieval in Mice: All animal procedures were approved by the Massachusetts Institute of Technology (MIT) Committee on Animal Care (Protocol no. 0620-045-023) and supervised by the MIT Division of Comparative Medicine veterinary staff. Female C57BL/6 mice of 8 weeks of age were obtained from Jackson Laboratory. Animals were anesthetized with 3% isoflurane in oxygen. To prepare for surgery, the surgical site was shaved and sterilized using alternating treatments of betadine and isopropanol. Non-medicated eye lubricant was applied using sterile cotton swabs to prevent corneal damage due to loss of blink reflex. Buprenorphine SR-LAB (0.03 mg; ZooPharm) and 1 ml of 0.9% sterile saline were administered subcutaneously for analgesia and hydration, respectively.

For intraperitoneal surgeries, a small incision (≈1 cm) was made along the midline of the abdomen, and a blunt dissection exposed the peritoneal lining. Another incision along the linea alba exposed the peritoneal cavity where implants were carefully inserted away from the fat pad. Simple interrupted stitches made using absorbable Vicryl sutures (Ethicon Inc.) closed the peritoneum incision. Wound clips closed the skin incision.

For subcutaneous surgeries, a small skin incision (≈1 cm) was made along the back of mice. Blunt dissection using a smooth-tip hemostat created a total of 4 pockets under the skin large enough to comfortably accommodate 8 mm diameter disks in the caudal and cranial locations on both sides of the spine. One implant was then inserted on each of the 4 locations before closing the incision using wound clips. Every implant in each mouse featured a distinct surface chemistry to account for animal-to-animal variability.

Prior to retrieval, mice were euthanized by CO_2 inhalation, followed by secondary euthanasia via cervical dislocation. Retrieval of fibrosed intraperitoneal implants and short-term (30 min) subcutaneous implants involved opening the cavity and locating implants. Fibrosed intraperitoneal implants had minimal attachment to surrounding tissue and were easily removed. However, fibrosed subcutaneous implants were intimately bonded to surrounding tissue, including the outer surface of the peritoneum and the inner surface of the skin. As such, retrieval required careful dissection to preserve the fidelity of pericapsular fibrotic tissue. All samples were transferred to pre-chilled sterile saline upon retrieval.

Quantitative Evaluation of Implant Fibrosis: Immediately following explantation, samples in saline were imaged on an optical microscope (Leica) for gross evaluation of implant fibrosis and then fixed in 10% formalin. Intraperitoneal implants were evaluated using a biochemical assay for total collagen content determination. A subset of these samples was set aside for qualitative assessment of connective tissue thickness and morphology by histology. Subcutaneous implants were evaluated by histology since substantial quantities of non-fibrotic tissue adhered to the fibrotic capsule would confound the biochemical assay.

Fibrosed intraperitoneal implants were sealed in glass vials containing 6 M HCl using Teflon thread seal tape to prevent evaporation. Boiling samples at 120 °C for at least 3 h hydrolyzed the collagen. The resulting solution was used for collagen quantification via a hydroxyproline assay (Chondrex, 6017; or Sigma, MAK008). Precise hydrolyzed solution volumes and implant surface areas were used to calculate the areal density of surface collagen for each sample.

Samples designated for histology were embedded in 2 wt.% low-melting point agarose to reinforce the fibrotic tissue in preparation for histological processing. Histology was performed at the Hope Babette Tang Histology Facility at the Koch Institute at MIT. Samples were dehydrated in a series of graded ethanol changes from 70% to 100% before clearing with xylene. During paraffin embedding, care was taken to orient the samples such that longitudinal cross sections of the implant could be obtained. Embedded samples were sectioned using a microtome, stained using Hematoxylin and Eosin (H&E) or Masson's trichrome stain, and scanned with a digital pathology slide scanner (Leica). For quantification of subcutaneous samples, stained sections were analyzed using ImageJ software to quantify the average fibrotic tissue thickness. Fibrotic tissue was distinguished by tightly packed, aligned collagen fibers (stained pink) and an increased density of nucleated cells (stained purple), including elongated fibroblasts, granulocytes, and foreign body giant cells. For each

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sample, at least 15 discrete measurements were made along the entire circumference of the fibrotic capsule and then averaged.

Immunohistochemistry began by dewaxing unstained histological sections in xylene and then rehydrating through a graded series of ethanol changes into distilled water. Samples were then permeabilized using 0.1% Triton X-100 in water for 2 h before treatment in a blocking buffer consisting of 1% bovine serum albumin, 0.1% Triton X100, and 0.05% Tween-20 in PBS for 2 h. Primary antibody staining against iNOS (Thermo Fisher Scientific, 14-5920-82), Arg1 (Cell Signaling Technologies, 93 668), and α-SMA (Sigma–Aldrich, ABT1487) was performed overnight at a 1:100 dilution in blocking buffer at 4 °C. Secondary antibody staining using anti-rat (488 nm) (Abcam, ab98420) or anti-rabbit (594 nm) (Vector Laboratories, DI-1594-1.5) fluorescently conjugated antibodies occurred at a 1:100 dilution in blocking buffer for 2 h in the dark. Thorough washes in tris-buffered saline (pH 7.4) were performed in between each step of the staining procedure. Autofluorescence quenching (10 min), DAPI nuclear counterstaining, and coverslip mounting were performed using a commercial kit (Vector Laboratories, SP-8400-15). Stained samples were imaged using a TissueFAXS SL fluorescent slide scanner (TissueGnostics) at 20x magnification along the entire long edge of each sample, then analyzed using the accompanying viewer software to quantify cell populations.

In Vitro Biomolecular Adsorption Studies: Fluorescein isothiocyanate (FITC) labeled albumin (Sigma–Aldrich, A9771) and 1-palmitoyl-2-(dipyrrometheneboron difluoride)undecanoyl-sn-glycero-3-phosphocholine (Avanti Polar Lipids, TopFluor PC, 810 281) were selected to assess the protein and phospholipid antifouling properties of coated PDMS respectively. Fouling solutions of FITC-Albumin were prepared at 1 mg mL^{-1} in deionized water and solutions of TopFluor PC were made at 0.5 mg mL^{-1} in deionized water. PDMS disks were incubated in fouling solutions for 1 h with shaking at 300 rpm. FITC-Albumin treated samples were washed three times with water and TopFluor PC treated samples were washed six times with water. Excess liquid was removed by lightly dabbing on a tissue wiper before air drying and imaging with a fluorescence microscope (Keyence BZ-X810, excitation 470/440 nm, emission 525/550 nm). Integrated fluorescence intensities from the resulting images were quantified on ImageJ. The relative intensities reported represent fluorescence measurements relative to those of uncoated PDMS control samples freshly prepared on the same day.

In Vivo Fouling Evaluation by Immunofluorescence Confocal Microscopy: Short-term, subcutaneous implantation followed the surgical procedure detailed in the previous section. After implanting 4 different samples in the dorsal subcutaneous space, mice were maintained under isoflurane-based anesthesia for 30 min before retrieval of samples into ice cold DPBS. Samples were then fixed for 30 min in 10% neutral buffered formalin for 30 min. Samples were then rinsed twice with PBS, then incubated with 0.5% bovine serum albumin and 0.05% sodium azide for 5 min before rinsing again with PBS and storing at 4 °C until further processing. Permeabilization and nuclear staining resulted from incubating samples in 0.1% Triton X-100 and 5 µg mL^{-1} of Hoechst 33 342 (Thermo Fisher Scientific, 62 249) in PBS for 30 min in the dark. Rinsing three times with PBS removed residual surfactant and dye. Blocking was performed by incubating samples in animal-free blocking agent (Vector Laboratories, SP-5030) for 30 min followed by two washes in PBS. Immunofluorescence staining resulted from incubating samples for 2 h in PBS containing fluorophore-conjugated antibodies targeting IgG (Host: Goat, Alexa Fluor 594, 1:500 dilution, Cell Signaling Technologies, catalogue no. 8890) and fibronectin (Host: Rabbit, Alexa Fluor 647, 1:75 dilution, Abcam, catalogue no. ab237287).

Stained samples were mounted on 6-well plates fitted with glass cover slip (Cellvis, P06-1.5H-N) and imaged on an Olympus FV1200 laser scanning confocal microscope. Due to large variability in fluorescence intensity across samples, imaging settings were individually adjusted for each sample to resolve characteristic surface features. Image analysis involved preprocessing images using the scikit-image (Version 0.22.0) library in Python (Version 3.11.1) before manually thresholding images in ImageJ to determine either cell density from stained nuclei counts or protein surface coverage. The preprocessing protocol facilitated easier and more accurate thresholding by applying a contrast-limited adaptive histogram equalization (CLAHE) to improve image contrast, followed by a Gaussian filter to blur high-frequency regions. Images were then converted to 8-bit files for image analysis. For protein coverage measurement, images simply received a binary threshold followed by area quantification. For nuclei counting, pre-processed images received further manual contrast adjustment since CLAHE can amplify crosstalk from other channels in the Hoechst 33 342 channel. Watershed filter separated cell clusters, and particle analysis (Size > 4 µm^2, Circularity = 0.6-1.0) defined the number of cells. Normalization by the area of each image gave the cell density. A total area of 3.23 mm^2 was analyzed for each implant and measurements across all locations were averaged to give a single data point per implant.

Primary Macrophage Isolation and In Vitro Culture: Bone marrow-derived macrophages were obtained from a female C57BL/6 mouse following a previously reported protocol.[87] After euthanasia, the skin was removed and the femurs and tibiae from the hind limbs were separated from the carcass into a sterile container kept on ice. The remaining muscles and tendons were then scraped away from the bones and the bones were sterilized by immersing in 70% ethanol for 30 s and thoroughly rinsed with DPBS. An 18 G needle was used to puncture a hole at the base of a 0.5 mL microtube before nesting the tube within a 1.5 mL microtube. Using sterile scissors and forceps, each bone was cut in half and placed cut-side-down within the 0.5 mL microtube. Centrifugation at 10 000 RCF for 1 min at 4 °C led to the bone marrow depositing at the base of the 1.5 mL microtube. The bone marrow was then resuspended in 500 uL of Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum, 1% penicillin streptomycin, and 50 µM β-mercaptoethanol. Filtering through a 70 µm strainer removed clumps in the cell suspension. The bone marrow cells were then cultured in supplemented DMEM with the addition of 20 ng mL^{-1} of recombinant mouse macrophage colony-stimulating factor (M-CSF) to induce macrophage differentiation. Media was changed after 3 days of culture (5% CO_2, 37 °C) and the resulting macrophages were used after a total of 7 days of differentiation.

For biomaterial response experiments, 1 cm diameter PDMS samples were prepared to ensure proper coverage of wells within a 48-well plate. PDMS disks were first rinsed with DPBS and fitted into the base of each well using sterile forceps, being careful to avoid scratching the PDMS surfaces. Before harvesting, macrophages were rinsed with DPBS three times to remove loosely adhered cells within the cell culture flask. 1×10^5 Macrophages were then plated into each well with 500 µL of supplemented DMEM with 20 ng mL^{-1} of M-CSF. Cells were incubated on surfaces for 2 days before cytokine and transcriptomic profiling.

Cytokine Release Quantification: After 2 days of culture, 450 µL of supernatant was extracted from each well and placed in a microtube. Centrifugation at 3000 RCF at 4 °C removed debris. 200 µL of this clarified supernatant was then stored at −80 °C until further analysis.

After thawing, Luminex xMAP technology enabled multiplexed quantification of mouse cytokines, chemokines and growth factors relevant to inflammation. The multiplexing analysis was performed using the Luminex 200 system by Eve Technologies Corp. (Calgary, Alberta). Ten markers were simultaneously measured in the samples using Eve Technologies' Mouse Focused 10-Plex Discovery Assay (MilliporeSigma) according to the manufacturer's protocol. The 10-plex consisted of GM-CSF, IFNγ, IL-1β, IL-2, IL-4, IL-6, IL-10, IL-12p70, MCP-1, and TNFα. Assay sensitivities of these markers range from 0.4 – 10.9 pg mL^{-1} for the 10-plex.

RNA Sequencing and Data Preparation: RNA was isolated from cultured macrophages on implant substrates using RNeasy Plus kit (Qiagen, 74 134) following the manufacturer's guidelines. For each sample, RNA was eluted in 30 uL of RNase-free water. The concentration was then measured using Nanodrop before storage at −80 °C prior to sequencing.

NEBNext Ultra II (New England Biolabs) Directional RNA preparation with poly(A) selection enabled high-throughput library preparation. Fragment size and quality of each sample was then confirmed on a 5300 Fragment Analyzer (Agilent) before sequencing on a Singular Genomics G4 sequencer using a 50 base pairs paired-end configuration.

RNA-seq data was used to quantify transcripts from the mm10 mouse assembly with the Ensembl version 110 annotation using the nf-core/rnaseq workflow revision 3.14.0.[88] Gene level summaries were prepared from the star_salmon quantitation using tximport version 1.28.0[89] running under R version 4.3.1 with tidyverse version 2.0.0. Differential

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expression analysis was done with DESeq2 version 1.40.2 using apeglm log fold change shrinkage.[90]

Gene Expression Analysis: Uniform manifold approximation and projection, similarity index calculations, and individual gene analysis was performed using Python (Version 3.11.1). To obtain a similarity index, the Euclidean pairwise distances between each sample was computed based on the transcript frequency of every gene, treating each gene as an individual dimension. The average distances between samples grouped by treatment group were then determined. Taking the inverse of the average distances and normalizing by the maximum value gave a normalized similarity index.

Specifically, a data table with n samples and p genes was used, where each gene value is a transcript frequency in units of transcripts per million. First, a distance matrix D of size $n \times n$ was generated, where the element D_{ij} represents the Euclidean pairwise distance between two samples x_i and x_j (where x_i and x_j are p-dimensional vectors representing the gene expression profiles of the i-th and j-th samples, respectively). The Euclidean distance $d(x_i,x_j)$ is given by:

$$d\left(x_i, x_j\right) = \sqrt{\sum_{k=1}^{p}\left(x_i - x_j\right)^2} \tag{1}$$

Using the distance matrix D, we calculated the average distance between treatment groups $\bar{d}(G_i, G_j)$, where G_i and G_j are two different treatment groups:

$$\bar{d}\left(G_i, G_j\right) = \frac{1}{|G_i| \cdot |G_i|}\sum_{s \in G_i}\sum_{t \in G_j} D_{st} \tag{2}$$

Here, $|G_i| \cdot |G_i|$ is the total number of pairwise distances considered between the two groups and D_{st} is the Euclidean distance between sample s in group G_i and sample t in group G_j. Taking the inverse average distance gave a measure of similarity $S(G_i,G_j)$ between two groups. Finally, normalization by the maximum value of $S(G_i,G_j)$ across all group pairs afforded a normalized similarity index.

Gene set enrichment analysis (GSEA) (version 2023.2.Mm) was performed using R to compare transcriptomes of macrophages cultured on antifibrotic surfaces against those cultured on untreated PDMS.[91] A significance threshold of false discovery rate of $q < 0.05$ was used. Network plots display significant gene sets as nodes colored according to normalized enrichment score (NES) and sized according to the number of individual genes within the set. Edges connect any two nodes (i.e., gene sets) whose constituent genes have significant (i.e., Jaccard index > 0.1) overlap. For gene sets A and B each consisting of some number of individual genes, the Jaccard index J is as follows:

$$J\left(A, B\right) = \frac{|A \cap B|}{|A \cap B|} \tag{3}$$

Nodes without any significant edges were omitted from visualizations for clarity.

Statistics: Details regarding sample size and appropriate statistical tests are described in the figure captions. Dunnett's multiple comparison tests were computed using the SciPy (Version 1.14.0) statistical functions module in Python (Version 3.11.1). Linear regressions and hypothesis testing using unpaired, two-tailed t test, or one-way analysis of variance (ANOVA) coupled with the Holm-Bonferroni method for multiple comparison correction were performed using OriginPro 2023. Statistical significance markers follow: NS = not significant, $*P < 0.05$, $**P < 0.01$, and $***P < 0.001$.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

D.G.A. and R.L. are founding scientists of Sigilon Therapeutics. For a list of entities with which R.L. is involved, compensated or uncompensated, see: https://www.dropbox.com/s/yc3xqb5s8s94v7x/Rev%20Langer%20COI.pdf?dl=0.

Author Contributions

W.J.J., and B.M.W. contributed equally to this work. W.J.J. and B.M.W. conceived the study, designed the experiments, and drafted the manuscript. W.J.J., B.M.W., Y.Z., R.M., A.L.J., J.G.R., S.L., and P.M. carried out the experiments and analyzed the results. W.J.J. and E.C. conducted the bioinformatics analysis. W.J.J., S.B., D.G.A. and R.L. supervised and secured funding for the study. All authors discussed and provided comments on the manuscript.

Data Availability Statement

All data needed to evaluate the conclusions in this paper are present in the paper and/or the Supporting Information.

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antifouling, biomaterials, coatings, fibrosis, immune response, implants, polymers

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